

DataPulse: Automated EDA Platform

Exploratory Data Analysis Made Simple

What is DataPulse?

DataPulse is a web-based automated exploratory data analysis (EDA) tool that I built to simplify the entire data analysis work low. It helps anyone - from beginners to professionals - understand their data quickly without writing a single line of code.

The Problem I Solved

When I started learning data science, I noticed that:

- Beginners struggle with writing repetitive EDA code
- Manual data cleaning is time-consuming and error-prone
- Visualizing data requires remembering complex syntax
- Getting actionable insights from data takes hours

My Solution: A single platform where you upload data and get everything - cleaning, visualization, machine learning, and insights - automatically.

What Can DataPulse Do?

1. Smart Data Upload

- Supports CSV and Excel files
- Instant preview of your data
- Automatic session management

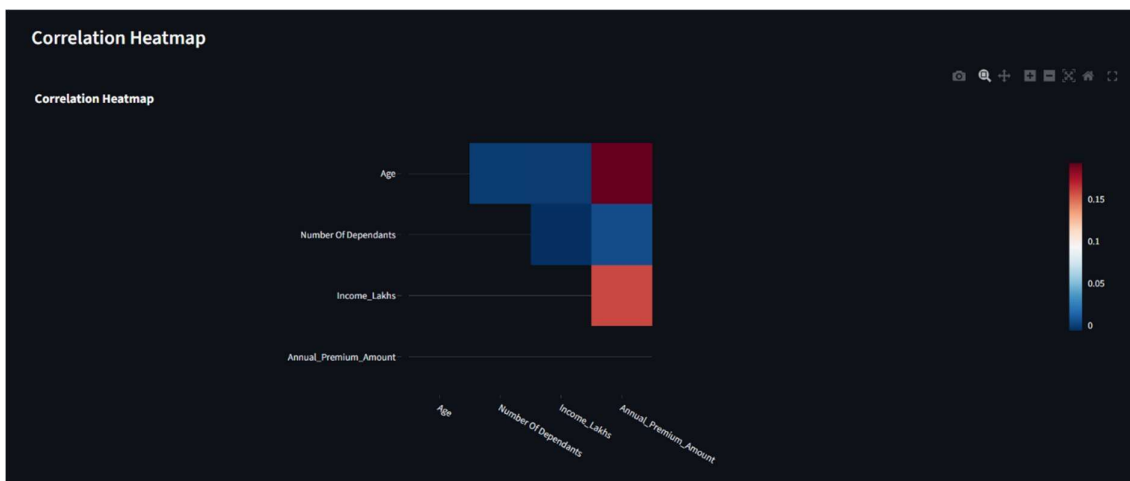
Summary	Manual Cleaning	Auto Cleaning	EDA	Outliers	Machine Learning	Insights
Dataset Summary						
Column Information						
	Column Name	Data Type	Missing Values	Unique Values		
0	Age	int64	0	59		
1	Gender	object	0	2		
2	Region	object	0	4		
3	Marital_status	object	0	2		
4	Physical_Activity	object	0	3		
5	Stress_Level	object	0	3		
6	Number Of Dependents	int64	0	9		
7	BMI_Category	object	0	4		
8	Smoking_Status	object	9	3		
9	Employment_Status	object	7	3		
Total Rows: 10000						
Total Columns: 15						
Duplicate Rows: 0						

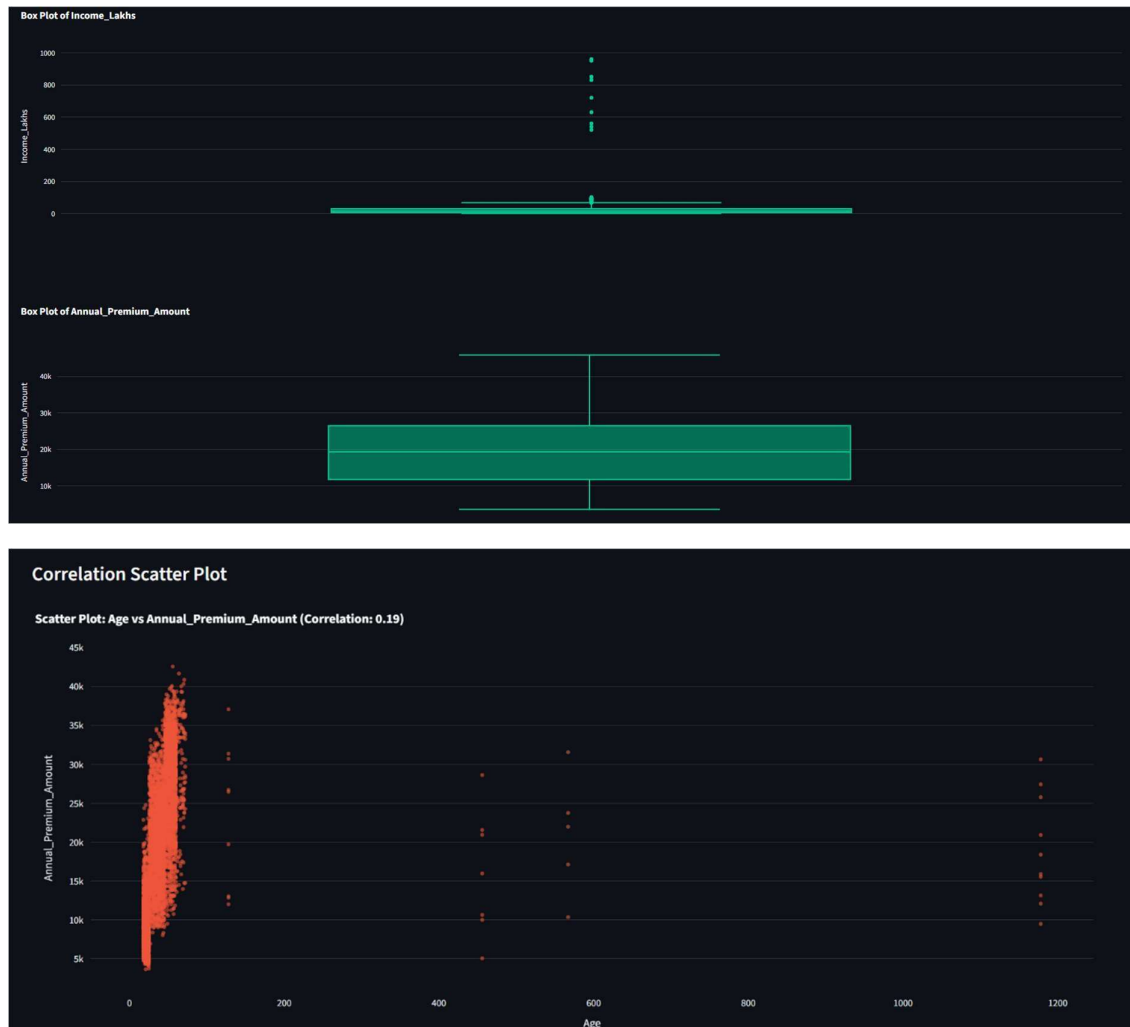
2. Intelligent Data Cleaning

- Manual Mode: Choose your own strategies for each column
- Auto Mode: AI-powered cleaning with one click
 - Handles missing values intelligently
 - Removes duplicates
 - Caps outliers using IQR method

3. Interactive Visualizations

- Distribution histograms
- Correlation heatmaps
- Scatter plots
- Box plots for outlier detection
- All charts are interactive (zoom, pan, hover)





4. Machine Learning

- Classification & Regression models
- 5 algorithms for each task
- Automated preprocessing
- Hyperparameter tuning option
- Performance metrics & visualizations

5. Professional Insights

- Comprehensive analysis reports
- Data quality assessment
- Actionable recommendations
- Business impact analysis

Technical Architecture

Frontend & Interface

- Streamlit: Web application framework
- Plotly: Interactive visualizations
- Custom CSS: Enhanced UI/UX

Data Processing

- Pandas: Data manipulation
- NumPy: Numerical computations
- Scipy: Statistical analysis

Machine Learning

- Scikit-learn: ML algorithms & preprocessing
- XGBoost: Advanced gradient boosting
- StandardScaler: Feature scaling
- OneHotEncoder: Categorical encoding

Advanced Features

- Google Gemini API: Enhanced insights generation
- Logging: Error tracking and debugging

Project Structure

DataPulse/

```
|— interface.py      # Main UI
|— functions.py      # Core logic
|— machinelearning.py # ML pipeline
|— llm_insights.py   # Insights engine
└— requirements.txt  # Dependencies
```

Feature Deep Dive - Data Cleaning

Why Data Cleaning Matters

Real-world data is messy:

- 80% of data science time is spent cleaning data
- Missing values can skew analysis
- Duplicates waste computational resources
- Outliers can mislead models

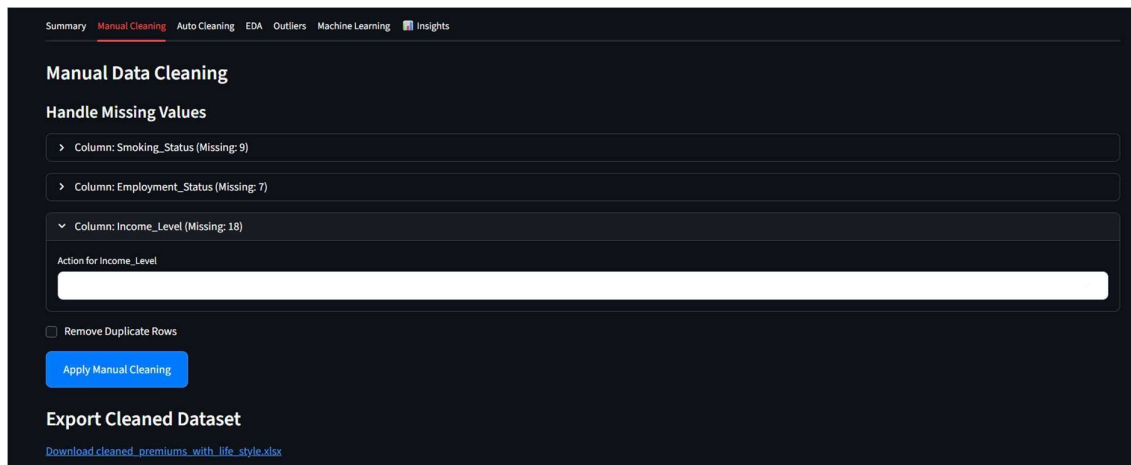
My Solution: Dual Cleaning Approach

Manual Cleaning

User Control: Choose strategies per column

- Drop rows with missing values
- Fill with Mean (for numeric data)
- Fill with Median (robust to outliers) ☐ Fill with Mode (for categorical data)
- Remove duplicates with checkbox

When to Use: When you understand your data and want specific control



The screenshot shows a web-based interface for manual data cleaning. At the top, there is a navigation bar with tabs: Summary, Manual Cleaning (active), Auto Cleaning, EDA, Outliers, Machine Learning, and Insights. The main heading is 'Manual Data Cleaning'. Below it, the section 'Handle Missing Values' is active. It lists three columns with missing values: 'Smoking_Status (Missing: 9)', 'Employment_Status (Missing: 7)', and 'Income_Level (Missing: 18)'. The 'Income_Level' column is expanded, showing a dropdown menu for 'Action for Income_Level' with a text input field. Below this, there is a checkbox for 'Remove Duplicate Rows'. A blue button labeled 'Apply Manual Cleaning' is at the bottom of the section. Below the button, the section 'Export Cleaned Dataset' is visible, with a link to 'Download cleaned_premiums_with_life_style.xlsx'.

Auto Cleaning

One-Click Solution: Intelligent automation

- Numeric columns → Fill with mean
- Categorical columns → Fill with mode
- Automatic duplicate removal
- Outlier capping using IQR method ($Q1 - 1.5 \times IQR$ to $Q3 + 1.5 \times IQR$)

When to Use: Quick analysis or when unfamiliar with data

Dataset Preview

	Age	Gender	Region	Marital_status	Physical_Activity	Stress_Level	Number Of Dependants	BMI_Category	Smoking_Status	Employment_Status	Income_Level	Income_Lakhs	Medical History	Insurance
0	56	Female	Southeast	Unmarried	Medium	Medium	2	Normal	No Smoking	Self-Employed	25L - 40L	37	High blood pressure	Gold
1	38	Female	Northeast	Married	Low	High	5	Overweight	Occasional	Freelancer	10L - 25L	10	No Disease	Silver
2	18	Female	Southwest	Unmarried	Medium	High	0	Normal	No Smoking	Freelancer	<10L	9	No Disease	Bronze
3	55	Male	Northeast	Married	High	Medium	3	Overweight	Occasional	Salaried	<10L	9	Thyroid	Silver
4	22	Male	Southeast	Unmarried	Medium	High	2	Normal	Regular	Self-Employed	10L - 25L	13	High blood pressure & Heart disease	Bronze

Summary Manual Cleaning **Auto Cleaning** EDA Outliers Machine Learning Insights

Automated Data Cleaning

Perform Auto Cleaning

Export Cleaned Dataset

[Download cleaned_premiums_with_life_style.xlsx](#)

Results Shown

- Cleaning report with details
- Before/after comparison
- Number of changes made

Page 6: Feature Deep Dive - Machine Learning

ML Pipeline Architecture

Step 1: Preprocessing

- Categorical Encoding: OneHotEncoder for text columns
- Numerical Scaling: StandardScaler for numeric features
- Missing Value Handling: Automated imputation
- Train-Test Split: 80-20 ratio
- Step 2: Model Training Classification Models:
- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost
- Support Vector Classifier

Regression Models:

- Linear Regression

- Decision Tree Regressor
- Random Forest Regressor
- XGBoost Regressor
- Support Vector Regressor

Step 3: Evaluation

- Classification: Accuracy, F1-Score, Confusion Matrix, ROC-AUC
- Regression: MSE, MAE, R² Score
- Cross-Validation: 3-fold CV for reliability
- Feature Importance: Which features matter most

Hyperparameter Tuning

- Optional GridSearchCV integration
- Optimizes model parameters automatically
- 3-fold cross-validation
- Takes longer but improves accuracy

Machine Learning

Select Task Type

Select Target Column

Select Model

☐ Enable Hyperparameter Tuning (Slower)

Selected Options: Task Type = Classification, Target Column = Gender, Model = LogisticRegression, Tuning = Off

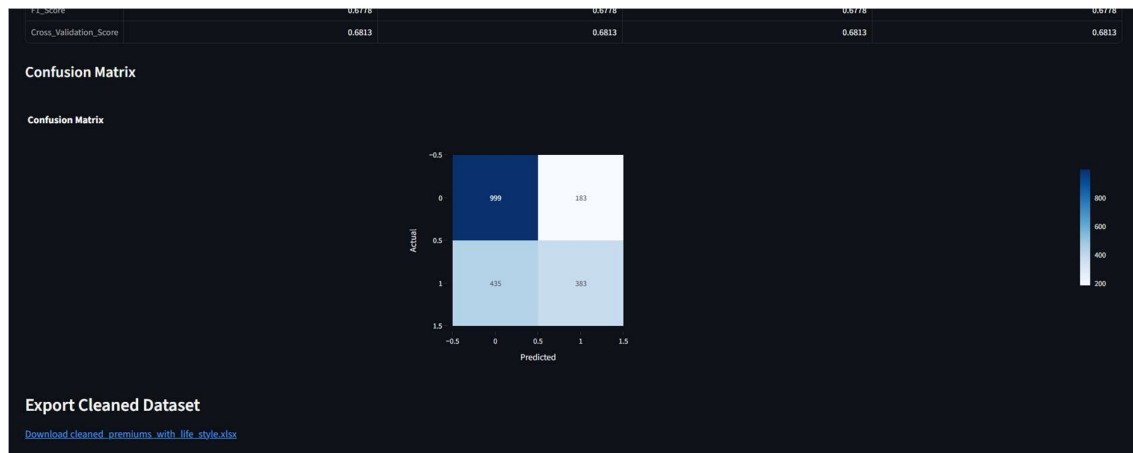
Train Model

Model trained successfully!

Model Evaluation

Classification Report:

	precision	recall	f1-score	support	
Female	0.6967		0.8452	0.7638	1182
Male		0.6767	0.4682	0.5535	818
accuracy		0.691	0.691	0.691	0.691
macro avg	0.6867		0.6567	0.6586	2000



Real-World Example

"For a customer churn dataset, the Random Forest classifier achieved 87% accuracy with cross-validation score of 0.85, identifying 'contract_type' and 'monthly_charges' as top predictive features."

Page 7: Feature Deep Dive - Insights Generation

The Insights Engine

How It Works

1. Statistical Analysis: My code generates traditional insights
 - o Missing value patterns
 - o Correlation strength
 - o Distribution characteristics
 - o Outlier detection
2. Enhancement Layer: Transforms raw stats into professional reports
 - o Dataset overview
 - o Key findings with explanations
 - o Data quality assessment
 - o Strategic recommendations
 - o Business impact analysis
3. Output: Comprehensive, easy-to-understand report
4. Two Modes

Enhanced Insights (Recommended)

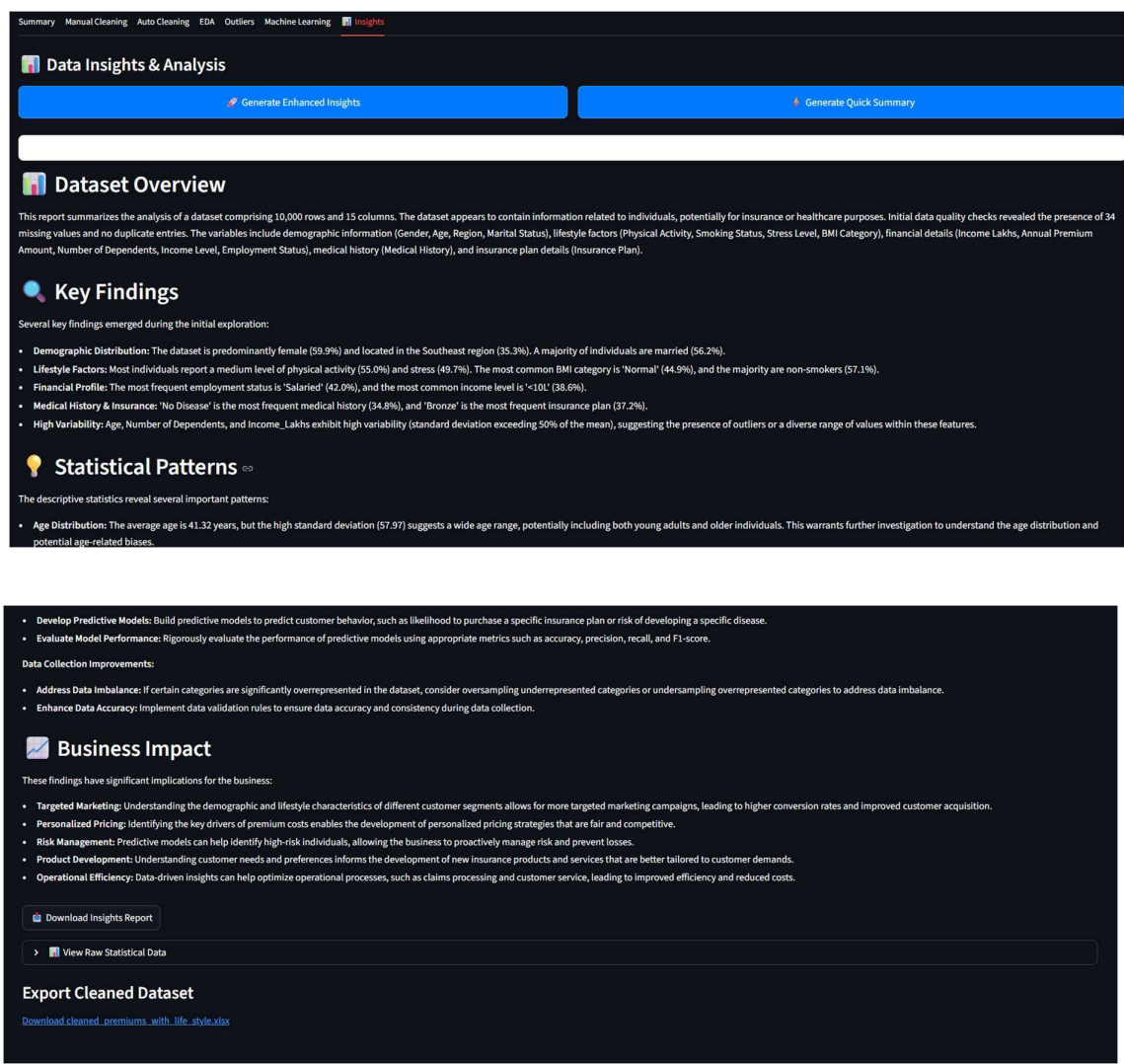
- Comprehensive analysis (6 sections)
- Professional formatting
- Actionable recommendations
- Takes 10-20 seconds

Quick Summary

- Fast 5-point analysis
- Instant results
- Good for quick checks

What Makes It Special

- Context-Aware: Considers cleaning and ML results
- Token-Optimized: Only sends summaries, not raw data (saves cost)
- Fallback Safety: Shows traditional insights if enhancement fails
- Downloadable: Export as text ile



Example Insight

Instead of: "Column A has correlation 0.85 with Column B"

It explains: "A strong positive correlation (0.85) exists between Column A and Column B, indicating these variables move together. This suggests potential redundancy in your features and could benefit it from dimensionality reduction techniques like PCA."

Example Use Case: Retail Store

Scenario: Store owner wants to understand sales patterns Using

DataPulse:

1. Upload sales CSV (product, quantity, price, date)
2. Auto-clean: Handle 5% missing values, remove 12 duplicates
3. EDA: Discover "electronics" correlates with weekend sales
4. ML: Random Forest predicts demand with 83% accuracy
5. Insights: "Weekend electronics promotions recommended"

Time Saved: 4 hours of coding → 10 minutes with DataPulse

- Rakesh Kapilavayi